

WELCOME TO

SITE / LEASE NAME

RIG #

KNOWN H₂S LOCATION

SAFETY IS MANDATORY — GO HOME SAFE

EMERGENCY CONTACTS

EMERGENCY DIAL
911

CANON SAFETY 24/7
(432) 556-3057

HOSPITAL — SOUTH OF MIDLAND RIGS
Midland Memorial (432) 221-1111

HOSPITAL — JAL, NM AREA RIGS
Lea Regional, Hobbs (575) 492-5000

XSTREMEMD 24/7 TELEMEDICINE
(877) 520-2911

911 ADDRESS — THIS LOCATION
.....

COMPANY MAN
.....

RADIO CHANNEL
.....

H₂S

H₂S IS COLORLESS, ODORLESS, AND DEADLY.
YOU CAN'T SEE IT. YOU CAN'T SMELL IT. IT CAN KILL YOU.

ENTRANCE FLAG — CHECK EVERY ENTRY

GREEN

H₂S POSSIBLE — NORMAL OPERATIONS

YELLOW

H₂S DETECTED — ESSENTIAL PERSONNEL ONLY

RED

DANGEROUS RELEASE — EVACUATE & MUSTER

SITE OVERVIEW

ATTACH THE ANNOTATED AERIAL / GATE-VIEW PHOTO OF THIS LOCATION
— OR SKETCH THE PAD LAYOUT HERE WITH A MARKER —
MARK MUSTER POINTS, WIND SOCKS, COMPANY MAN TRAILER & ACCESS ROADS

GREEN MUSTER — UPWIND

RED MUSTER — CROSSWIND

COMPANY MAN TRAILER

WIND SOCKS

ACCESS ROADS

KEEP ACCESS ROADS CLEAR AT ALL TIMES FOR EMERGENCY VEHICLES

GREEN MUSTER POINT
HEAD INTO THE WIND — WIND IN YOUR FACE

RED MUSTER POINT
UPWIND BLOCKED? CUT ACROSS THE WIND

WHEN IN DOUBT — HEAD INTO THE WIND

SITE INFORMATION & EMERGENCY

01 CONTACTS

COMPLETE AT RIG-UP ·
VERIFY EVERY CREW
CHANGE

PLAN REVISION DATE PREPARED PREPARED BY REVIEW DUE

LOCATION

OPERATOR

DRILLING CONTRACTOR & RIG NO.

WELL NAME & NUMBER

API NUMBER

COUNTY / STATE

GPS (LAT / LONG)

911 PHYSICAL ADDRESS / DIRECTIONS FROM NEAREST HIGHWAY

RADIO CHANNEL

MUSTER POINTS (PRIMARY / SECONDARY)

ROLE

NAME

PHONE

Company Man (Operator rep)

Toolpusher / Rig Manager

Canon Safety technician on site

Canon Safety 24/7 dispatch

Operator drilling superintendent

Hospital — south of Midland rigs

Midland Memorial Hospital · 400 Rosalind
Redfern Grover Pkwy, Midland, TX

(432) 221-1111

Hospital — Jal, NM area rigs

Lea Regional Medical Center · 5419 N
Lovington Hwy, Hobbs, NM

(575) 492-5000

XstremeMD 24/7 telemedicine

Remote medics — call on any injury or
exposure

(877) 520-2911

Air ambulance / medevac

EMERGENCY NOTIFICATION ORDER

- 1 ALERT THE CREW.** Sound the alarm / call it out on the radio — channel above. Everyone to muster.
- 2 COMPANY MAN TAKES COMMAND.** Accounts for all personnel against the gate log.
- 3 LIFE THREAT OR INJURY — CALL 911.** Give the 911 physical address exactly as written above.
- 4 CALL XSTREMEMD 24/7 — (877) 520-2911.** Remote medics direct first aid until EMS arrives. Then Canon Safety 24/7 — (432) 556-3057.
- 5 COMPANY MAN NOTIFIES THE OPERATOR'S DRILLING SUPERINTENDENT.** No re-entry until the all-clear.

02 KNOW THE HAZARD

WHAT H2S IS ·
WHAT IT DOES TO YOU

COLORLESS & INVISIBLE

You cannot see it — in air, in the cellar, over the pits.

KILLS YOUR SENSE OF SMELL

Rotten-egg odor at low levels, then it deadens your nose. No smell ≠ no gas.

HEAVIER THAN AIR

Collects in cellars, pits, tanks and low spots — the places you climb into.

FLAMMABLE

Ignites at 4.3% in air and burns to sulfur dioxide — also toxic.

CARRIED BY WELL FLUIDS

Released from mud returns, drilling breaks, connections, trips and swabbing.

FAST-ACTING

High concentrations drop you in one or two breaths — no warning.

WHERE IT SHOWS UP ON A DRILLING RIG

Shale shakers & mud returns · the cellar & substructure · mud pits & tanks · drilling breaks & kicks · connections & trips · flowline & possum belly · any closed or low-lying space

LEVEL (PPM)	WHAT IT MEANS FOR YOU
< 1	Rotten-egg smell may be noticeable. Odor is not a safe gauge — treat any smell as a detection.
10	Low alarm. Eye and throat irritation with time. Leave the area, report the reading.
15	High alarm. Emergency response — muster upwind, account for everyone.
100	Immediately dangerous to life & health (IDLH). Kills your sense of smell in minutes.
500+	Staggering, collapse and respiratory failure within minutes.
1,000	Immediate knockdown — unconscious in one or two breaths. Death in minutes without rescue.

Reference: NIOSH ceiling 10 ppm (10-min) · OSHA ceiling 20 ppm · IDLH 100 ppm. Personal monitor setpoints per operator standard — see page 4.

EARLY SYMPTOMS — GET OUT & REPORT

Eye burning & watering · headache · dizziness · nausea · coughing / sore throat · unusual fatigue

SERIOUS — RESCUE RESPONSE NOW

Loss of smell · confusion · shortness of breath · staggering · collapse ("knockdown") · unconsciousness

Symptoms can be delayed — report every suspected exposure, even if you feel fine.



NEVER RELY ON YOUR SENSE OF SMELL

H₂S CAN KILL WITHOUT WARNING — YOUR MONITOR IS THE ONLY THING THAT TELLS THE TRUTH

03 ALARMS & WHAT TO DO

KNOW THE SETPOINTS ·
MOVE WITHOUT HESITATION

10

PPM · LOW

LOW ALARM — LEAVE THE AREA

Stop the task, move upwind, and report the reading to the Company Man before anyone goes back to work.

15

PPM · HIGH

HIGH ALARM — EMERGENCY

Full alarm response below — muster, account for everyone, and hold for the Company Man's all-clear.

IF THE ALARM SOUNDS

1

STOP WORK IMMEDIATELY

2

PUT ON ESCAPE HOOD (IF AVAILABLE)

Don't lose time searching — moving to clean air comes first.

3

MOVE UPWIND / CROSSWIND TO YOUR MUSTER POINT

Look at the wind sock and head into the wind. Never run downwind. Never go low.

4

ACCOUNT FOR ALL PERSONNEL

Company Man checks the count against the gate log.

5

DO NOT RE-ENTER UNTIL THE COMPANY MAN GIVES THE ALL-CLEAR

Air must be verified clear by monitor — not by time passed.

WIND SOCKS — KNOW YOUR SAFE AREA

H₂S is **heavier than air and travels with the wind** — downwind and low ground is where the gas goes. Clean air is **into the wind**: look at the sock and put the wind in your face.

WALK INTO THE WIND



The sock's tail points downwind, where the gas goes — you go the opposite way

GREEN — UPWIND →

UPWIND BLOCKED BY THE RIG OR THE CLOUD?



Cut 90° across the wind to clear it, then work upwind

RED — CROSSWIND →

WHEN IN DOUBT — HEAD INTO THE WIND

ENTRANCE CONDITION FLAGS

GREEN

H₂S possible. Normal operations — monitors on, stay aware.

YELLOW

H₂S detected above threshold. Non-essential personnel leave; masks staged.

RED

Dangerous release. Evacuate to muster. SCBA only for response crew.

The Company Man sets the entrance flag as site conditions change — check it every time you pass the gate.

ACCOUNTABILITY & ALL-CLEAR

Everyone on location signs the gate log — that log **is** the muster count. The Company Man (or designee) is the **only** person who gives the all-clear, and only after the air is verified clear by monitor. Missing person? Tell the Company Man immediately — **do not go back in on your own.**

04 PROTECTION & DETECTION

REQUIRED AT ALL TIMES ·
NO EXCEPTIONS

PERSONAL PROTECTION — REQUIRED AT ALL TIMES



WEAR YOUR PERSONAL H₂S
MONITOR AT ALL TIMES



H₂S AWARENESS TRAINING
CARD ON YOUR PERSON



HARD HAT · SAFETY GLASSES ·
GLOVES · FR CLOTHING ·
SAFETY-TOE BOOTS



SUPPLIED AIR / SCBA STAGED
ON SITE FOR EMERGENCY USE

IF YOU ARE NOT TRAINED OR PROPERLY EQUIPPED — DO NOT ENTER

YOUR MONITOR — EVERY SHIFT

- ☐ Worn in the **breathing zone** (collar / lapel — not your belt or bag)
- ☐ **Turned on** before you pass the gate, on until you leave
- ☐ **Bump tested** per schedule and within **calibration** date
- ☐ Alarm you can **hear and feel** over rig noise (audible + vibrate)
- ☐ Any alarm or reading gets **reported** — never silence and carry on

SUPPLIED AIR ON THIS SITE

ESCAPE PACKS STAGED AT

SCBA UNITS LOCATED AT

CASCADE / AIR TRAILER

- Escape packs are for **getting out** — never for entry or work.
- SCBA entry is for **trained, fit-tested responders only**.
- A respirator can't seal over facial hair — keep the seal area clean-shaven.

FIXED DETECTION ON THIS RIG

BELL NIPPLE / FLOWLINE SENSOR SHAKER AREA SENSOR

MUD PIT SENSORS CELLAR / SUBSTRUCTURE SENSOR

ALARM SIGNAL (HORN / STROBE) CHECKED BY / FREQUENCY

Mark each fixed sensor on the site overview (page 1). Canon Safety verifies sensor and alarm function on every service visit.

WORKING RULE

Nobody works alone in a known H₂S area. Use the **buddy system**, keep line of sight or radio contact, and know where your nearest escape pack and muster point are **before** the job starts — not when the alarm sounds.

05 WHO DOES WHAT & RESCUE

KNOW YOUR ROLE ·
BEFORE YOU NEED IT

COMPANY MAN (OPERATOR REP)

- Commands the response; accounts for all personnel
- The only person who gives the **all-clear** and authorizes re-entry
- Makes external notifications (911, operator, Canon 24/7)

TOOLPUSHER / RIG MANAGER

- Backs up command; directs rig crew at muster
- Coordinates well-securing with the Driller
- Confirms contractors & third parties are accounted for

DRILLER

- Sounds the alarm on confirmed detection
- Secures the well per the drilling program before leaving the floor — if safe to do so
- Musters the floor crew; reports count to Toolpusher

DERRICKMAN & FLOOR CREW

- Execute shutdown tasks assigned by the Driller
- Move upwind/crosswind to muster; buddy check on the way
- Report anything unusual — smells, readings, symptoms

CANON SAFETY TECHNICIAN

- Continuous air monitoring; maintains detection & air equipment
- Runs gate control, briefings and the sign-in log
- Supports rescue with SCBA, cascade air & medical aid

VISITORS & THIRD PARTIES

- Check in at the company man trailer before entering the pad
- Complete the pre-entry checklist (page 7); stay with your escort
- On alarm: muster with your escort — do not freelance

H₂S EXPOSURE — RESCUE RESPONSE

- 1 DO NOT BECOME A VICTIM.** Never enter an H₂S atmosphere without proper training, supplied air and authorization — most H₂S multiple fatalities are would-be rescuers.
- 2 ALERT.** Notify others and call for help on the radio.
- 3 REMOVE.** If safe to do so — on air — move the victim upwind to fresh air.
- 4 ASSESS.** Check breathing and responsiveness.
- 5 RESPOND.** Basic first aid within your training only — no medical intervention beyond it. XtremeMD and EMS direct all care.
- 6 SUMMON.** Call 911, then XtremeMD 24/7 — (877) 520-2911 — and Canon Safety (432) 556-3057.
- 7 REPORT.** Report the incident to the Company Man immediately.

AFTER ANY EXPOSURE OR ALARM EVENT

- **Medical evaluation for anyone exposed — even if they feel fine.** H₂S effects, including fluid on the lungs, can show up hours later. Start with XtremeMD — (877) 520-2911.
- Preserve monitor data logs. Note the time, location, wind direction and who was where.
- Company Man reports to the operator; Canon Safety completes the incident report and follow-up.

MEDICAL ON SITE — BASIC FIRST AID ONLY

FIRST AID KIT LOCATED AT

No medical intervention happens on location. For any injury or exposure: call **911**, then **XtremeMD 24/7 — (877) 520-2911** — remote medics direct first aid until EMS arrives.

NEAREST HOSPITALS — CIRCLE THIS SITE'S

South of Midland: Midland Memorial Hospital — **(432) 221-1111**
400 Rosalind Redfern Grover Pkwy, Midland, TX

Jal, NM area: Lea Regional Medical Center — **(575) 492-5000**
5419 N Lovington Hwy, Hobbs, NM

Post the printed route map beside this plan.

06 GATE GUARD — PRE-ENTRY CHECKLIST

VERIFY WITH EVERY PERSON ·
EVERY TIME

☐ Do you have your H ₂ S awareness training card ?	☐ Do you understand the alarm response and muster location ?
☐ Do you have a personal H₂S monitor on you — turned on?	☐ Have you been briefed by the Company Man ?
☐ Is your monitor bump tested and within calibration ?	☐ Do you know your job, your location, and who you're working for ?
☐ Are you wearing proper PPE ? (hard hat, safety glasses, gloves, FR clothing, safety-toe boots)	☐ Do you have a radio — or know the channel your escort is on?
☐ Do you understand this is a known H₂S location ?	☐ Are you feeling well and fit for duty today?

IF ANY ANSWER IS NO — DO NOT ENTER. STOP AND INFORM THE COMPANY MAN.

DATE	NAME	COMPANY	TIME IN	TIME OUT	MONITOR #	BRIEFED BY

This gate log is the muster count during an emergency — keep it accurate and current. Photocopy this page as needed; leave the master in the plan.

07 H2S SAFETY MEETING & SIGN-IN

RUN THE MEETING STRAIGHT
OFF THIS PLAN — PAGES 1-7

DATE TIME RIG / WELL

CONDUCTED BY MEETING TYPE

BRIEFING AGENDA — COVER EVERY ITEM				
☐ Known H2S location — site layout, muster points, wind socks	P.1	☐ Emergency contacts, radio channel & 911 address	P.2	
☐ What H2S does — exposure levels, no-smell rule	P.3	☐ Alarm setpoints & the five alarm-response steps	P.4	
☐ Wind sock reading — green upwind / red crosswind	P.4	☐ Monitors — worn high, on, bump tested, in calibration	P.5	
☐ PPE & supplied-air locations (escape packs, SCBA, cascade)	P.5	☐ Roles — who commands, who secures the well, who gives all-clear	P.6	
☐ Rescue — do not become a victim; the seven steps	P.6	☐ Crew questions answered · everyone fit for duty	—	

NAME (PRINT)	COMPANY	H2S CERT CURRENT	MONITOR ON & BUMPED	SIGNATURE
		☐	☐	
		☐	☐	
		☐	☐	
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Photocopy this page for each meeting — file completed sheets with the rig's HSE records. Drills: walk the crew to both muster points on the first tour and after any rig move.